

RATNESH KUMAR SINGH

Electronics & Communication Engineer | VLSI |

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LinkedIn: <https://tinyurl.com/2st86aht>

Portfolio: <https://my-portfolio-omega-liat-40.vercel.app/>

GitHub: <https://github.com/Ratnesh919>



PROFILE SUMMARY

Electronics and Communication Engineering graduate with a focused interest in VLSI, digital electronics, and communication systems. Hands-on exposure to Verilog HDL and embedded hardware development. Currently learning SystemVerilog for design verification.

SKILLS

Programming

C, C++, Python, Verilog, HTML

Embedded

Arduino, sensors

Tools

EDA Playground, VS Code, GitHub, Tinkercad

PROJECT

Smart Antenna For Vehicular Application Prototype

Nov 2025 – Jul 2026

- Modeled and optimized a 536 MHz compact monopole antenna in ANSYS HFSS for V2V/V2X communication links.
- Attained excellent return loss (-31.87 dB), 1.05 VSWR, and 98.3% radiation efficiency across a 55 MHz operating bandwidth).
- Reduced antenna height by 74% (from 14 cm to 3.6 cm) using coaxial ceramic dielectric loading and capacitive end-loading.

ACADEMIC QUALIFICATIONS

Qualification	Institution	Board / University	Year
B.Tech (ECE)	Swami Vivekananda Institute of Science & Technology	MAKAUT (W.B.U.T.)	2022–2026
12th — Science (PCM)	P.B.S College, Bihar	B.S.E.B	2018–2020
10th — Science	Vidyanaganj High School (H.S.)	I.G.C.S.E	2017–2018

CERTIFICATIONS & TRAINING

- Python Programming — 1-month certification focused on Python scripting and algorithms.
- C Language Programming — 1-month certification focused on programming fundamentals.
- EV Service Technician — 1-month program covering EV electronics, diagnostics, and hardware testing.
- GIS (Geographic Information Systems) — 2-week training course.

INTERESTS

- Practicing Verilog HDL design concepts using EDA Playground.
- Learning System Verilog for design verification
- Tracking AI trends

WORKSHOPS & LANGUAGES

Workshops HAM Radio , BSNL industrial visit, Cybersecurity awareness.
Languages English, Hindi , Bengali